

5.1.11 Properties of Summations and Products

The following theorem states general properties of summations and products. The proof of the theorem is discussed in Section 5.6.

Theorem 5.1.1

If $a_m, a_{m+1}, a_{m+2}, \dots$ and $b_m, b_{m+1}, b_{m+2}, \dots$ are sequences of real numbers and c is any real number, then the following equations hold for any integer $n \geq m$:

1. $\sum_{k=m}^n a_k + \sum_{k=m}^n b_k = \sum_{k=m}^n (a_k + b_k)$
2. $c \times \sum_{k=m}^n a_k = \sum_{k=m}^n c \times a_k$ generalized distributive law
3. $\left(\prod_{k=m}^n a_k \right) \times \left(\prod_{k=m}^n b_k \right) = \prod_{k=m}^n (a_k \times b_k)$